

INDIVIDUAL CONTRIBUTION ESSAY

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Course: DAMO-611

(Project: Machine Learning for Credit Default Prediction)

1. What I Did to Help the Group Accomplish the Deliverable

My contributions to the final course project focused on four major areas that were essential to the successful execution of the group deliverable:

1. **Topic Selection and Dataset Acquisition** – I sourced, evaluated, and proposed the project topic as well as the dataset we ultimately used. I identified the “Default of Credit Card Clients” dataset from the UCI Machine Learning Repository and presented it to my group, where it was reviewed and accepted as the foundation of our project.
2. **Development of the Introduction Section** – I took full responsibility for crafting the introduction portion of the report, which included the motivation, purpose, scope, research questions, hypotheses, and methodology. This section set the foundation and direction for the entire project.
3. **Coding and Technical Contribution** – I actively participated in writing the Python code used for data exploration, statistical testing, preprocessing, and model development. This included contributing to the implementation of Logistic Regression and Random Forest models, along with assisting in running t-tests, chi-square tests, and initial descriptive analysis.
4. **Report Writing and Refinement** – In addition to the introduction, I supported the team in drafting and refining other parts of the final report, ensuring that our findings were clearly communicated, aligned with course objectives, and professionally presented.

Together, these contributions helped the group maintain momentum and complete the deliverables to a high standard.

2. How I Did It

I approached my responsibilities strategically and collaboratively:

- **Sourcing the Dataset:**
I explored multiple reputable data sources and evaluated potential datasets for quality, relevance, completeness, and alignment with course expectations. After selecting the UCI credit card dataset, I shared its structure, advantages, and feasibility with the group to initiate project planning.
- **Creating the Introduction Section:**
I conducted background research on credit default prediction, reviewed academic literature, and synthesized information into a clear and academically sound

introduction. I ensured that the introduction aligned with our modeling objectives and supported the flow of the entire report.

- **Coding Contribution:**

I worked closely with teammates during coding sessions. I wrote and tested portions of the code in Python, including handling imbalance through class weights, helping structure the feature selection process, and validating statistical tests. I also helped debug issues and verified that outputs were consistent with our research goals.

- **Report Collaboration:**

Throughout the writing process, I helped refine tone, clarity, formatting, and alignment between sections. I assisted in transforming code results into understandable insights and ensuring the narrative was cohesive across all group members' contributions.

3. Analysis of What I Did

My contributions were foundational to both the **direction** and **execution** of the project:

- **Impact of Topic & Dataset Selection:**

By identifying a real-world credit default dataset with strong predictive modeling potential, I enabled the group to work on a problem that was both academically rigorous and practically relevant. This early contribution shaped the entire project trajectory, ensuring feasibility and depth.

- **Importance of the Introduction Section:**

The introduction I developed provided the conceptual and methodological framework for the rest of the project. It allowed all team members to work with clarity regarding the purpose, research questions, hypotheses, and overall project boundaries. This significantly improved our group's coordination and productivity.

- **Effectiveness of My Coding Contribution:**

My involvement in coding strengthened the analytical quality of the project. I helped ensure that statistical tests were correctly implemented, preprocessing was handled appropriately, and the models ran smoothly. This improved the validity of our results and supported accurate interpretation.

- **Contribution to Collaboration:**

I consistently supported the team through communication, initiative, and willingness to assist wherever needed. My role acted as a bridge between conceptual planning and technical execution, helping maintain cohesion across different parts of the project.

Overall, my contributions were essential in both the foundation and analytical depth of the final deliverable.